ABSTRACT

This study evaluates :using satellite-derived irradiation data from the PVGIS-SARAH3 database spanning 2005–2023. In the absence of building-level solar assessment platforms in much of Sub-Saharan Africa, the study applies a dual modelling framework combining Fixed Effects panel regression with a within-transformed XGBoost machine learning model to estimate monthly irradiation on optimally inclined rooftop surfaces. Results reveal strong seasonal variability, a statistically significant negative temperature effect consistent with photovoltaic efficiency theory, and a modest but positive long-term trend in rooftop solar potential. City-level comparisons indicate that Tamale exhibits the highest potential despite thermal constraints, while Kumasi records the lowest due to persistent cloud cover. Accra and Tema emerge as priority locations for urban rooftop solar deployment given their favorable irradiation and demographic importance. The study demonstrates the value of integrating satellite data, econometric methods, and machine learning to support solar planning in data-scarce environments.

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1. INTRODUCTION

Globally, there has been an intensification of efforts towards renewable energy, especially solar power. (Hassan et al., 2024). The traditional approach to solar resource estimation often involves field-based surveys and ground-mounted instrumentation. These methods are costly, time-consuming, and unable to cover large spatial areas. Urban and peri-urban areas in the Global South have seen an increase in solar installations. (Hutchings et al., 2022). There is therefore the increase need to evaluate solar potential at regional and national scales with both speed and precision.

The integration of Geographic Information Systems (GIS) and Artificial Intelligence (AI) into geospatial technologies has been a transformative tool for evaluating the rate of energy transitions and for predictive models for policy intervention. (Vaiyapuri & Julie, 2025). Through automated analysis of satellite imagery, three-dimensional terrain models, and climate datasets, it is increasingly possible to estimate and predict solar irradiance and roof-level generation capacity across millions of structures. (Hissou et al., 2023). An example of such an automated system is Google Solar API, which leverages computer vision and Machine Learning (ML) to assess the solar potential of over 400 million buildings across 40 countries. Google Solar constructs detailed models of roof geometry, simulates shading from surrounding structures and vegetation, and incorporates localized microclimate data to produce site-specific solar energy estimates at an unprecedented scale. (Menezes et al., 2024). However, as of 2026, Google Solar is unavailable in many African countries where the high-resolution geospatial and building data required to power such systems either do not exist or are insufficiently comprehensive.

As a result, in sub-Saharan Africa, data gap impedes the training and validation of AI-driven solar estimation models as the region lacks a dense network of ground-based meteorological stations and high-resolution spatial datasets (Osei-Dwomoh & Forkuo, 2025). Data unavailability and accessibility issues have informed the use of indirect methods for assessing solar resources to advance solar energy planning and policy implementation in SSA (Borzì, 2025). This study uses the PVGIS-SARAH3 satellite-derived irradiation database as a proxy for estimating rooftop solar potential. PVGIS provides monthly irradiation data at location-specific resolution, thereby enabling city-level comparative analysis in regions where data from the Google Solar API is unavailable. This paper focuses on five major cities in Ghana: Accra, Kumasi, Tema, Takoradi, and Tamale. Ghana is a compelling case study for Global South countries in SSA for many and various reasons. First of all, the country occupies a high-irradiance position within the West African solar belt. The country is rapidly urbanizing, with the current urbanization rate pegged at almost 60% (Badasu et al., 2025). The Government of Ghana has also articulated a national renewable energy target that requires robust spatial planning support. However, systematic, spatially resolved assessments of rooftop solar potential across Ghanaian cities remain limited in the peer-reviewed literature.

This study makes three primary contributions. Firstly, it applies a Fixed Effects panel regression to 19 years of monthly PVGIS irradiation data. The model properly accounts for the panel structure of city-level longitudinal data. Secondly, this paper uses a Machine Learning model, specifically a within-transformed XGBoost, to capture the nonlinear relationships that the linear specification cannot detect. Lastly, it derives city-level solar potential profiles that can directly inform renewable energy investment and policy planning in Ghana and comparable data-scarce environments.

2. LITERATURE REVIEW & THEORETICAL UNDERPINNINGS

This study is underpinned by three theories: the Resource Potential Assessment Framework (RPAF), the Photovoltaic Systems Theory, and the temperature coefficient theory.

2.1. Resource Potential Assessment Framework

The foundation of this study is the Resource Potential Assessment Framework, which provides a hierarchical taxonomy for evaluating energy resources across successive levels of constraint. RPAF distinguishes among the four nested levels of potential: theoretical, technical, economic, and market (Schaldach et al., 2026). The theoretical component of the RPAF can be understood in terms of the total solar radiation incident on a given surface area, which is constrained by the fundamental laws of physics and the geometry of solar-terrestrial interaction. It constitutes the upper bound of what is physically available and is independent of technological, economic, or social considerations. Technical potential narrows this estimate by accounting for the constraints imposed by available conversion technologies, including panel efficiency, system losses, roof area availability, and installation feasibility. Economic potential further restricts the technically recoverable resource to the subset that is financially viable under prevailing cost structures and energy prices (Angliviel de La Beaumelle et al., 2023). Market potential, at the base of the hierarchy, represents what can realistically be deployed given institutional, behavioral, and regulatory constraints.

This study operates at the theoretical-to-technical potential boundary. The target variable H(i_opt)_m monthly irradiation on the optimally inclined plane represents the theoretical maximum solar energy receivable by a rooftop photovoltaic system at each location, assuming optimal panel tilt and orientation. This metric most closely aligns with what building-level assessment platforms, such as the Google Solar API, compute, making it the most appropriate proxy for rooftop solar adoption potential in the absence of such tools. By focusing on this boundary, the study provides the foundational spatial evidence upon which technical, economic, and market assessments can subsequently be built.

The RPAF is particularly well-suited to the Ghanaian context because it accommodates the data constraints inherent in developing country settings. Where comprehensive building stock data, electricity tariff structures, and financing frameworks are unavailable or inconsistent, theoretical potential assessment using satellite-derived irradiation offers the most feasible and scientifically defensible entry point into the renewable energy planning hierarchy. The framework thus justifies the methodological choice to use PVGIS-SARAH3 irradiation data as a proxy for building-level solar resource assessment, positioning this study as an empirical contribution to the theoretical-technical potential estimation stage of Ghana's solar energy planning process.

The RPAF also provides the normative rationale for the city-level comparative analysis conducted in this study. Urban centers are the primary loci for realizing technical and market potential, given their concentration of building stock, electricity consumers, grid infrastructure, and institutional capacity. Ranking cities by theoretical potential as this study does for Accra, Kumasi, Tema, Takoradi, and Tamale directly informs the prioritization of technical feasibility studies and investment planning at the next stage of the RPAF hierarchy.

2.2. Photovoltaic Systems Theory

The second theoretical pillar of this study is drawn from photovoltaic systems theory, which provides the physical and engineering basis for understanding how solar irradiation is converted into electrical energy and the conditions under which this conversion efficiency varies. Three theoretical principles from this body of literature are directly operationalized in the analytical framework of this study. (Sang et al., 2025).

The first is the optimal tilt angle theory, grounded in the geometric relationship between the angle of solar incidence and the irradiance received by a tilted surface. Based on the work of Liu & Jordan (1960), on the angular distribution of solar radiation, it is well established that a photovoltaic panel-oriented perpendicular to the mean solar vector, that is, tilted at an angle approximately equal to the site latitude and facing the equator, maximizes the annual energy yield. This principle directly justifies the selection of H(i_opt)_m, the irradiation on the optimally inclined plane, as the dependent variable in this study. By targeting the optimally tilted surface irradiation rather than the horizontal irradiation H(h)_m, the study captures the maximum realizable rooftop solar potential rather than a conservative lower bound.

The second principle is the solar radiation decomposition theory, originally formalized by (Liu & Jordan, 1960) and subsequently extended by Erbs et al., (1982). This theory posits that global solar irradiation on a horizontal surface can be decomposed into direct beam and diffuse components, with the ratio of diffuse to global irradiation (Kd) serving as a measure of atmospheric transmissivity. Under clear-sky conditions, Kd is low, and direct-beam radiation dominates, resulting in high irradiation on tilted surfaces. Under overcast conditions, Kd is high, and diffuse radiation dominates, reducing both the total irradiation and the effectiveness of optimal tilting. This theoretical decomposition underpins both the inclusion of Kd as a predictor variable and the Erbs correlation model used to estimate Kd for cities where it was not directly available from the PVGIS download.

The third, and perhaps most policy-relevant, principle is the temperature-coefficient **theory** of photovoltaic performance degradation. Standard photovoltaic panels are characterized and rated under Standard Test Conditions (STC), defined as an irradiance of 1000 W/m², a cell temperature of 25°C, and an air mass of 1.5. In practice, cell temperatures routinely exceed 25°C under field conditions, particularly in tropical environments such as Ghana. The power output of a crystalline silicon photovoltaic panel declines at a rate of approximately 0.4 to 0.5 percent per degree Celsius above the STC reference temperature, a relationship described by the temperature power coefficient (γ) (Green, 1982; Skoplaki & Palyvos, 2009):

3. METHODOLOGY

3.1. Study Area

Five cities in Ghana: Accra, Tema, Kumasi, Takoradi, and Tamale were chosen for this study. The cities were chosen to ensure geographic, climatic, demographic, and policy representativeness across Ghana's major urban centers. This, therefore, allows the findings to be generalizable across other cities, especially in West Africa. Accra (Ghana's most populous city and national capital) and Tema are located on Ghana's southern coast, and together they form the country's administrative and industrial enclave. Takoradi also houses one of Ghana's important ports in the country and is the third most populous city in Ghana. Kumasi is Ghana's second most populous city and is located in the country's forest zone. Tamale is the largest city in northern Ghana, situated in the Guinea Savanna zone and with a markedly different climate from southern cities.

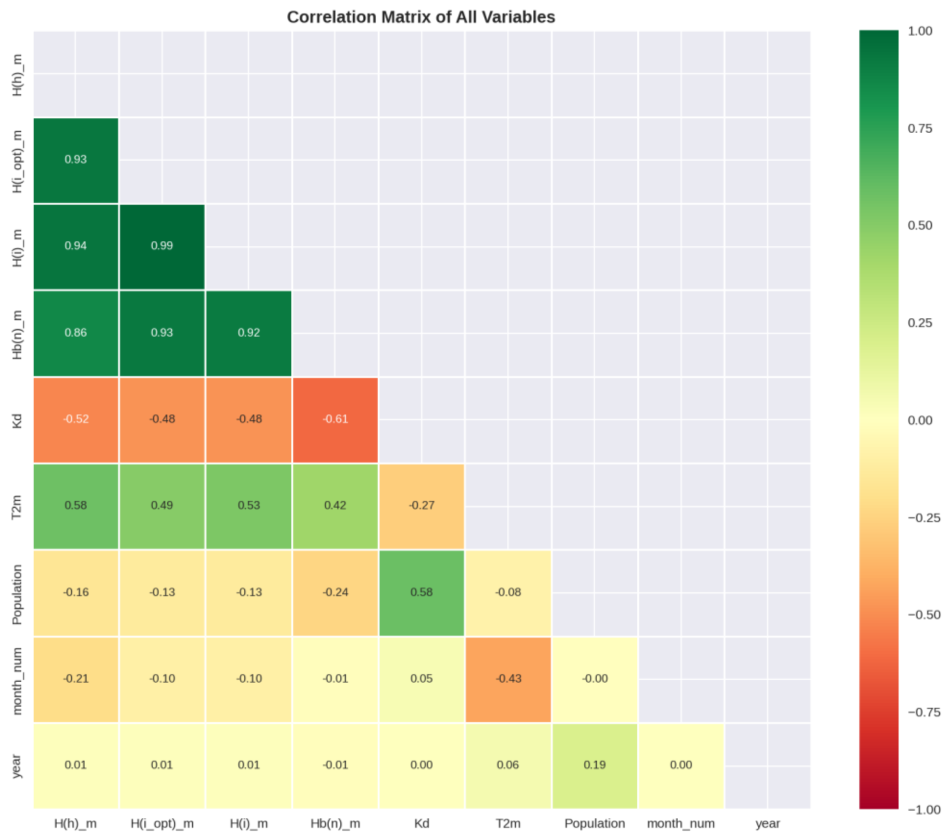
3.2.Solar Irradiation Data

3.3.The data for this study were acquired from the PVGIS-SARAH3 database for the period January 2005 to December 2023. Solar Irradiation Data and Population Data.

The data for this study were obtained from the PVGIS-SARAH3 database for the period from January 2005 to December 2023. This extracted data yielded 228 monthly observations per city. Six variables were taken from this database: global horizontal irradiation, irradiation on the optimally inclined plane, irradiation on a plane at 10° tilt, monthly beam irradiation on the normal plane, diffuse-to-global irradiation ratio, and, lastly, the 24-hour average temperature. The dependent variable (or target) was the irradiation on the optimally inclined plane. It was chosen as the target or dependent variable because it represents the maximum solar energy receivable by a rooftop photovoltaic system in each city and closely approximates the output that a building-level tool such as the Google Solar API would compute. A correlation matrix was computed (as shown in Figure 1), which indicates a near-perfect correlation with the target variable; therefore, this variable was dropped to prevent data leakage.

The population data, on the other hand, for the five cities under study were sourced from the Ghana Statistical Service from its 2000, 2010, and 2021 Population and Housing Censuses. Linear interpolation was used to impute population figures for years where data were missing.

Figure 1. Correlation Matrix of all variables



3.4. Models

Two models were used in this study. The first is the Fixed Effects model (FE). To select the FE, the study ran Pooled OLS, FE, and Random Effects models, and conducted a Hausman test to select the best method. The results showed that FE was best for this analysis. The second model used an ML model, specifically the within-transformed XGBoost. This approach was chosen because it is equivalent to the within estimator used in FE regression and ensures that the XGBoost model learns within-city temporal and seasonal variations rather than just between-city differences. For the ML model, the dataset was trained on data from 2005 - 2019, while the test set was evaluated on data from 2020 – 2023. The parameters used were 500 estimators, a learning rate of 0.05, a maximum depth of 6, a subsample ratio of 0.8, a minimum child weight of 5, an L1 regularization of 0.1, and an L2 regularization of 1. As this task was a regression problem, the model performance was evaluated using the root mean square error (RMSE) and the mean absolute error (MAE).

4. EXPLORATORY DATA ANALYSIS

Before the multivariate and ML analyses were conducted, an exploratory data analysis (EDA) was performed to examine the distributions, seasonal patterns, temporal trends, and intervariable correlations within the dataset. The target or dependent variable showed an approximately normal distribution, with a mean of 175.2 kWh/m²/month and no extreme outliers (as shown in Figure 2). In the city-specific analysis, Figure 2 shows that Kumasi exhibited a left-skewed distribution. It indicates a higher frequency of low-irradiation months. At the same time, Tamale exhibited a right-skewed distribution, consistent with its Savanna climate and a higher proportion of high-irradiation observations.

A seasonal analysis revealed a consistent bimodal pattern across the five cities under consideration. There was peak irradiation in January–March and November–December; these months represent the country's harmattan season. The remaining months showed a trough, which coincides with the country's rainy season. In the five cities observed, Tamale showed a marked difference in its seasonal profile, with substantially higher irradiation and recording its lowest value one month later than the coastal cities. This can be explained by the fact that Tamale experiences a longer harmattan and a shorter rainy season. Lastly, the annual trend analysis for the 2005 – 2023 period showed stable irradiation levels across all five cities.

Figure 2. Distribution of the target or dependent variable

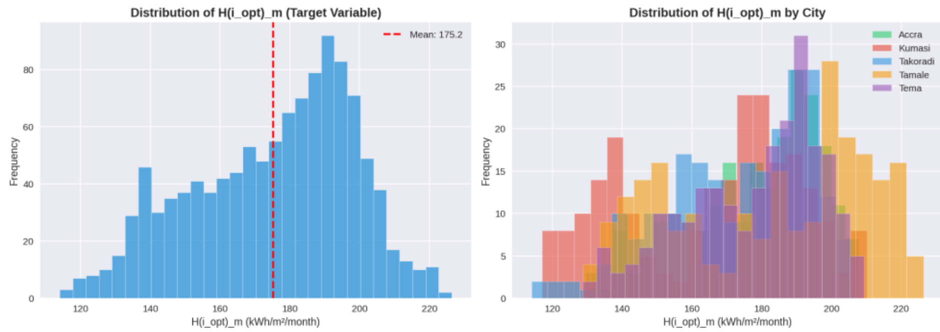


Figure 3. Average monthly solar irradiation by city

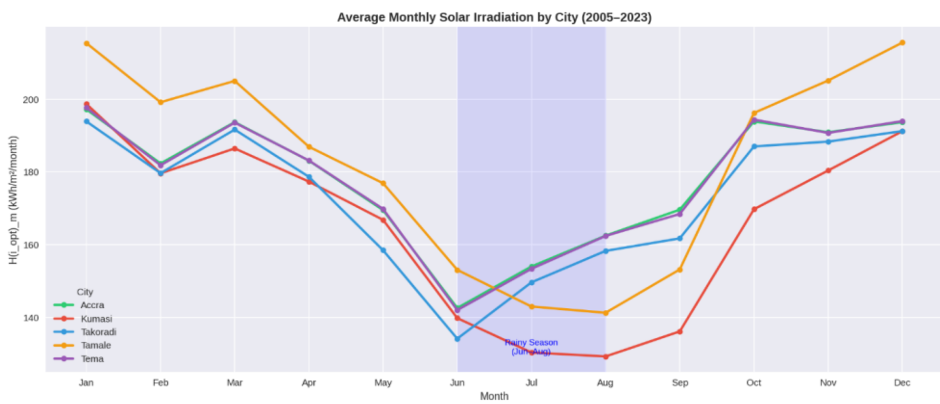


Figure 4. Average solar irradiation trend by city

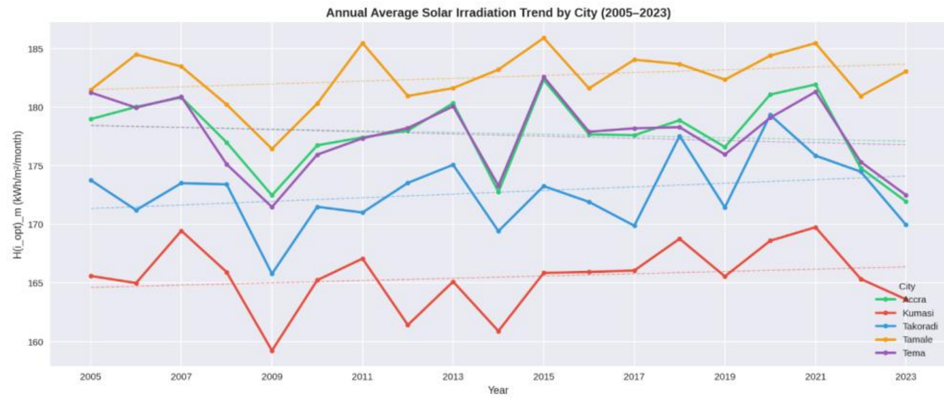
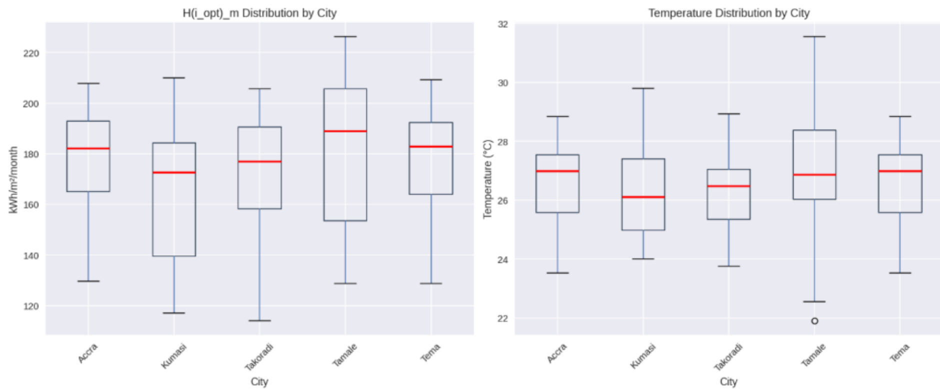


Figure 5. Temperature distribution by city



5. MULTIVARIATE AND ML RESULTS

5.1. RESULTS FROM THE FE MODEL

The FE model shows that Horizontal irradiation (H(h)_m) is the strongest statistically significant predictor ($\beta = 0.617$, $p < 0.001$). The model shows that for every 1 kWh/m² increase in horizontal irradiation within a city, optimal rooftop irradiation increases by 0.617 kWh/m². Direct beam irradiation (Hb(n)_m) contributes a significant additional positive effect ($\beta = 0.264$, $p < 0.001$), capturing the incremental value of clear-sky direct sunlight to rooftop panel performance. The model also shows that Temperature has a significant negative effect on optimal rooftop irradiation ($\beta = -1.000$, $p < 0.001$). This finding is theoretically consistent with the temperature coefficient phenomenon in photovoltaic physics, whereby panel efficiency declines at approximately 0.4 to 0.5 percent per degree Celsius above the standard test condition of 25°C. In Ghana's high-temperature environment, particularly in Tamale and during the dry harmattan season, thermal efficiency losses partially offset the irradiation advantage, underscoring the importance of temperature adjustment in realistic solar adoption assessments.

A statistically significant positive temporal trend was identified ($\beta = 0.054$, $p = 0.001$), suggesting that rooftop solar potential across Ghanaian cities has increased by approximately 0.054 kWh/m²/month per year over the study period. Cumulated over 19 years, this represents a total incremental gain of approximately 1.03 kWh/m²/month, consistent with gradual improvements in atmospheric transparency associated with reduced aerosol loading and land use changes.

The monthly dummy variables collectively reveal a pronounced seasonal structure. June recorded the deepest irradiation deficit relative to the January peak ($\beta = -20.05$, $p < 0.001$), consistent with Ghana's major rainy season driven by the northward advance of the Inter-Tropical Convergence Zone. March was statistically indistinguishable from January ($\beta = -0.075$, $p = 0.974$), confirming both months as peak solar periods. The irradiation recovery from the rainy season trough proceeds steadily from September through December, with near-peak conditions restored by December.

Table 1. Results for the FE model

Variable	Coefficient	Std. Error	T-statistic	P-value
Constant	-41.549	34.048	-1.220	0.223
H(h)_m	0.6166	0.0697	8.840	0.000***
Hb(n)_m	0.2644	0.0542	4.879	0.000***
Kd	-5.3947	6.0429	-0.893	0.372
T2m	-0.9999	0.1099	-9.095	0.000***
Population	0.0008	1.0928	-0.152	0.879

Variable	Coefficient	Std. Error	T-statistic	P-value
year	0.0543	0.0159	3.408	0.001***
February	-2.464	0.810	-3.043	0.002**
March	-0.075	2.267	-0.033	0.974
April	-11.170	1.197	-9.330	0.000***
May	-18.395	0.391	-47.084	0.000***
June	-20.051	0.160	-125.20	0.000***
July	-17.861	0.844	-21.166	0.000***
August	-13.275	1.480	-8.970	0.000***
September	-10.321	1.233	-8.368	0.000***
October	-6.411	1.065	-6.021	0.000***
November	-3.821	0.598	-6.391	0.000***
December	-0.632	0.209	-3.029	0.003***

5.2.ML RESULTS

5.2.1. Model evaluation

The model evaluation from the ML shows that, per city, Kumasi had the highest predictive accuracy despite the seasonal variability observed in the EDA, indicating that the model learned the deep rainy-season trough and dry-season peak structure characteristic of the transitional forest-savanna zone. These findings suggest that while the within-transformed XGBoost framework performs robustly across diverse geographic contexts, cities with distinctive localized atmospheric dynamics may benefit from supplementary location-specific features in future modelling efforts.

Table 2. Model Evaluation

City	R^2	RMSE	MAE	Performance rank
Kumasi	0.9968	1.4375	1.0979	1st
Tema	0.9921	1.6893	1.1990	2nd
Accra	0.9902	1.7990	1.2294	3rd
Tamale	0.9943	2.1320	1.5521	4th

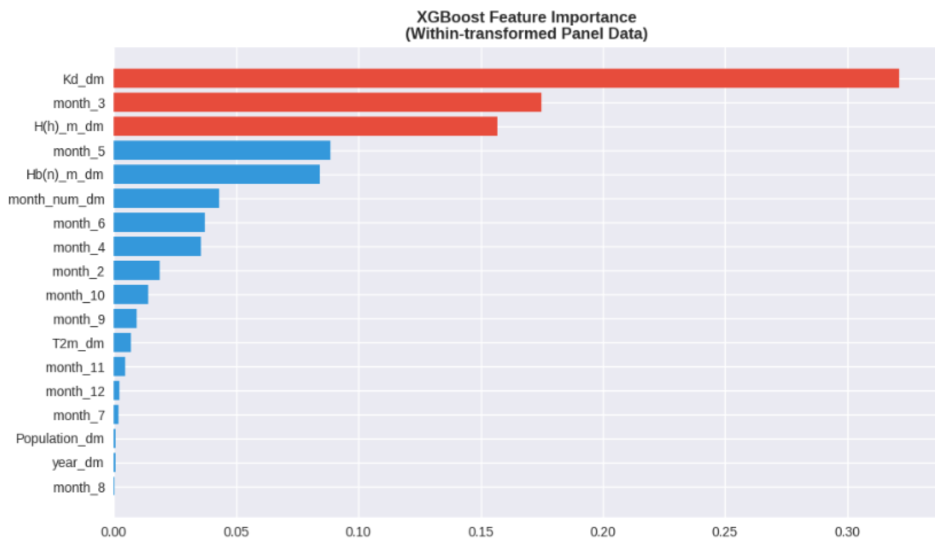
Takoradi	0.9887	2.1303	1.3557	5th
Overall	0.9936	1.8569	1.2868	

5.2.2. Results from the feature importance

The feature importance scores are presented in Figure 6. Based on the figure, it is apparent that the diffuse-to-global irradiation ratio (Kd) is the single most important predictor (accounting for 32.1%) of the total feature importance. However, as compared to the FE, this variable was not statistically significant. The divergence between the two models shows that Kd's relationship with rooftop solar potential is non-linear and context-dependent, operating through threshold and interaction effects that the linear panel specification cannot detect but that XGBoost captures naturally through its tree-splitting mechanism.

The third most important feature was horizontal irradiation (15.7%); this, however, is consistent with the FE specification. Direct beam irradiation (Hb(n)_m) ranked fifth at 8.4%. March seasonality (month_3) ranked second at 17.5%, reflecting the sharp transition from peak irradiation conditions in January–March to the onset of the rainy season, making March a critical inflection point in Ghana's solar calendar. Temperature (T2m) ranked eighth at 0.67%, substantially lower than its significance in the Fixed Effects model, suggesting that while temperature effects are statistically real, their magnitude relative to irradiation and cloud cover is modest in the machine learning framework.

Figure 6. Feature Importance of the various variables



5.3.Model Comparison

The results in Table 3 indicate that the two models are complementary rather than competing. While the FE model provides statistical inference and explainability, the ML model enables prediction and captures non-linear dynamics, especially the threshold behavior of cloud cover, making it a great tool for generating solar potential forecasts at a monthly resolution.

Table 3. Model Comparison

Metric	**Fixed Effects (V2)**	**XGBoost Within**
R^2 Within	0.9945	0.9932
R^2 Overall	0.9856	0.9936
RMSE (kWh/m²/month)	—	1.857
MAE (kWh/m²/month)	—	1.287
Handles non-linearity	No	Yes

Metric	Fixed Effects (V2)	XGBoost Within
Statistically interpretable	Yes	Partial (feature importance)
Hausman-validated	Yes	N/A
Recommended use	Inference & policy	Prediction & planning

6. DISCUSSION

6.1.Solar Potential across Ghanaian cities

The results affirm that Ghana has strong potential for solar energy, due to its location in the Guinea Savanna zone.(Stock et al., 2023). Ghana experiences lower cloud cover and stronger direct-beam radiation, which is more pronounced in southern cities. (Junior et al., 2022). However, Tamale's advantages are partially offset by its significantly higher temperature due to its location, indicating that location can negatively affect panel efficiency. Kumasi, located in the middle belt of the country, recorded the lowest average irradiation among the five cities. The city has a longer rainy season, which produces more prolonged cloud cover and higher diffuse irradiation ratios than any other city in the study. Accra and Tema, which are only 25km apart, showed nearly identical irradiation profiles and benefit from the relatively dry coastal microclimate of southeastern Ghana, where the effects of the Benguela current reduce cloud formation during the minor rainy season.

The message to policymakers is that, because Tema and Accra have over 4.5 million inhabitants, it is important to prioritize solar energy, as they have a high-density, moderate-to-high irradiation target area for rooftop solar deployment.

6.2.Temporal Trend and Climate implication

This paper lays the ground for Ghana to prioritize energy transition through solar energy planning. Over 19 years, the study found a cumulative gain of approximately 1.03 kWh/m²/month in rooftop solar potential across all cities. This cumulative gain affirms a study by Nikkath & Selvi (2024), which has attributed the increases in gradual irradiation to reductions in aerosol optical depth associated with improved air quality and changing land-use patterns. This means there is an economic advantage to investing in solar in Ghana and other countries in SSA. The results show that solar resource bases across Ghanaian cities are not deteriorating; therefore, they provide confidence in financial models underpinning solar energy investments with a 20–25-year lifespan.

6.3 Temperature as a Structural Constraint on Solar Adoption

The results show a significant negative temperature coefficient, which is often overlooked in high-level resource assessment. For instance, Tamale, which records higher average temperatures than other southern cities, has an irradiation advantage over Accra of approximately 2–4 kWh/m²/month due to thermal efficiency losses alone. This finding reinforces the case for incorporating temperature data into solar resource assessments for tropical environments and supports the use of temperature-corrected performance ratio calculations in investment appraisals.

7. CONCLUSION

This study provides a comprehensive assessment of rooftop solar energy potential in Ghana across five major cities. This study was done using 19 years of PVGIS-SARAH3 satellite irradiation data, panel econometric methods, and machine learning. The principal findings are as follows. The study shows that Tamale has the highest rooftop solar potential while Kumasi has the lowest. This has implications for energy economics. Tamale's solar potential can generate employment opportunities, provide a cheaper source of energy, and help reduce the barriers to rural-urban migration. Among the study sites, Kumasi has the lowest rooftop solar potential. Accra and Tema are strategically important targets for urban rooftop solar deployment because of their moderate-to-high irradiation, large populations, and status as Ghana's economic, administrative, and industrial hubs.

In SSA, where data is often unavailable, this study shows that employing PVGIS as a proxy for building-level solar assessment tools, as well as combining statistical and ML models, is helpful in obtaining results that are important for policy implementation. This offers transferable knowledge that other SSA countries can use. Future research should integrate building footprint data from OpenStreetMap to estimate actual rooftop areas, incorporate electricity demand profiles to compute solar self-sufficiency ratios, and extend the analysis to additional cities to map national-scale solar adoption potential.

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